

PARALLEL COMPUTING — CLASS 0 DIAGNOSTIC

Prerequisite & problem-solving diagnostic • 2 hours

Purpose: This session checks your current programming, algorithmic, data, and systems knowledge. You are not expected to know parallel computing yet. Work individually and show your reasoning.

Instructions: Use a language you know. Do not worry about perfect syntax unless asked. For algorithm questions, explain your thinking. If you do not know something, say so and move on.

Task 1 — Trace the Code (5 min)

Determine the final output. Show the value of x and y after each statement.

```
x = 5
y = 2
x = x + y
y = x * 2
x = y - x
print(x, y)
```

Answer:

- After `x = 5` : `x = 5` , `y = ?`
 - After `y = 2` : `x = 5` , `y = 2`
 - After `x = x + y` : `x = 7` , `y = 2`
 - After `y = x * 2` : `x = 7` , `y = 14`
 - After `x = y - x` : `x = 7` , `y = 14`
 - **Final Output** `print(x, y)` : **7 14**
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Task 2 — Find the Bug (5 min)

What does this program actually calculate? What was it probably intended to calculate? Fix it.

```
numbers = [10, 20, 30, 40, 50]
total = 0
for i in range(len(numbers)):
    total = numbers[i]
print(total)
```

Answer:

- **Actually calculates:** Prints only the last element (50) because `=` overwrites `total` .
 - **Intended to calculate:** Sum of all numbers ($10 + 20 + 30 + 40 + 50 = 150$).
 - **Fix:** Change `total = numbers[i]` to `total += numbers[i]` .
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Task 3 — Nested Loops (5 min)

For the program below: (a) how many times does `print()` execute? (b) what if both loops run from 0 to 999? (c) what if the outer loop runs 10 times and the inner loop runs 1,000,000 times?

```
for i in range(10):
    for j in range(10):
        print(i, j)
```

Answer:

- (a) **100** times (\$10 $\times$ 10\$).
 - (b) **1,000,000** times (\$1000 $\times$ 1000\$).
 - (c) **10,000,000** times (\$10 $\times$ 1,000,000\$).
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Part B — Algorithms & Complexity

Task 4 — Which Is Faster? (5 min)

Algorithm A scans an array once. Algorithm B compares every element with every element. For an array of size n , estimate the amount of work for each. Which grows faster? What happens when $n = 1,000,000$?

Answer:

- **Work:** Algorithm A = $O(n)$, Algorithm B = $O(n^2)$.
 - **Faster growth:** Algorithm B grows quadratically faster.
 - **When $n = 1,000,000$:** Algorithm A performs 10^6 ops (fast), Algorithm B performs 10^{12} ops (extremely slow).
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Task 5 — Complexity Ranking (7 min)

Rank these from fastest-growing to slowest-growing as n increases:

n^2 $\log_2 n$ n $n \log_2 n$ 1 2^n

Which become particularly problematic when n becomes much larger?

Answer:

- **Ranking:** $2^n > n^2 > n \log_2 n > n > \log_2 n > 1$.
 - **Problematic:** 2^n (uncomputable for $n > 50$) and n^2 (impractical for large n).
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Task 6 — Searching (7 min)

Given [3, 8, 12, 17, 24, 31, 45, 51, 63], find 45 using two approaches: (1) if you do not know the list is sorted; (2) if you know it is sorted. Explain why the second can be faster.

Answer:

- **(1) Unsorted:** Linear Search ($O(n)$) — check elements sequentially from start to end.
 - **(2) Sorted:** Binary Search ($O(\log_2 n)$) — check middle element and discard half the list.
 - **Why faster:** Binary search cuts the search space in half at every step.
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Part C — Data & Problem Solving

Task 7 — Large Dataset (8 min)

You have 500 million integers and need to find the largest number. Design an algorithm. How many times does each number need to be examined? Do you need to store all 500 million numbers? Could you process the data in chunks?

Answer:

- **Algorithm:** Stream numbers one by one, keeping a running maximum `max_val = max(max_val, current)` .
 - **Examinations:** Exactly **1 time per number** ($O(n)$ total).
 - **Store all: No**, only 1 current number needs to be kept in memory ($O(1)$ space).
 - **Chunks: Yes**, process chunks in parallel across threads, then take the maximum of local maxima.
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Task 8 — Counting (5 min)

Given A B A C B A D C A B, design an algorithm producing A->4, B->3, C->2, D->1. What data structure would you use?

Answer:

- **Data Structure:** Hash Map (Dictionary).
 - **Algorithm:** Iterate sequence, increment frequency count for each character in Hash Map.
 - **Output:** A->4, B->3, C->2, D->1 .
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Task 9 — Matrix Operations (5 min)

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. Calculate $A+B$ and $A*B$. How many individual multiplication operations are required for two 2×2 matrices?

Answer:

- **$A + B$:** $\begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$
 - **$A \times B$:** $\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$
 - **Multiplication operations:** 8 scalar multiplications.
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Part D — Systems Intuition

Task 10 — CPU vs Memory (6 min)

A program scans a huge array, but the CPU is extremely fast and the program is still slow. Give three possible reasons.

```
for x in huge_array: total += x
```

Task 11 — What Happens When You Run a Program? (7 min)

Describe what happens between typing `./program` and seeing the program's output. Use ordinary language. Include anything you think is relevant.

Task 12 — Four Programs (7 min)

Four programs run simultaneously: A uses lots of CPU; B uses lots of memory; C reads a large file; D mostly waits for user input. Identify the likely bottleneck for each. Can all four run at the same time? What determines CPU time? What changes if there is one CPU core?

Part E — Reasoning Challenges

Task 13 — The Slow Program (6 min)

A program processes 1 billion records in 10 hours. A student says: "If we buy a computer that is twice as fast, the program will take 5 hours." Do you agree? Explain.

Task 14 — Make It Faster (6 min)

A program takes 60 seconds. You may change anything except the answer it produces. List as many ways as possible to reduce execution time.

Task 15 — Bottleneck Identification (6 min)

Execution times are: A=2s, B=2s, C=50s, D=3s, E=2s. Total=59s. If A, B, D and E become 10× faster, how much faster does the entire program become? Where should optimization effort go?

Final Challenge — Design the Algorithm

Task 16 — 10 Million Student Marks (10 min)

You receive 10 million student marks, each between 0 and 100. You must produce: minimum, maximum, average, median, number scoring ≥ 90 , and number scoring < 40 . Design an algorithm; identify data structures; estimate how many times the data must be examined; estimate memory requirements; identify the likely bottleneck; and explain how you would make it faster without changing the answer.

STOP HERE. Check that you attempted every task. You are not expected to know parallel computing yet.